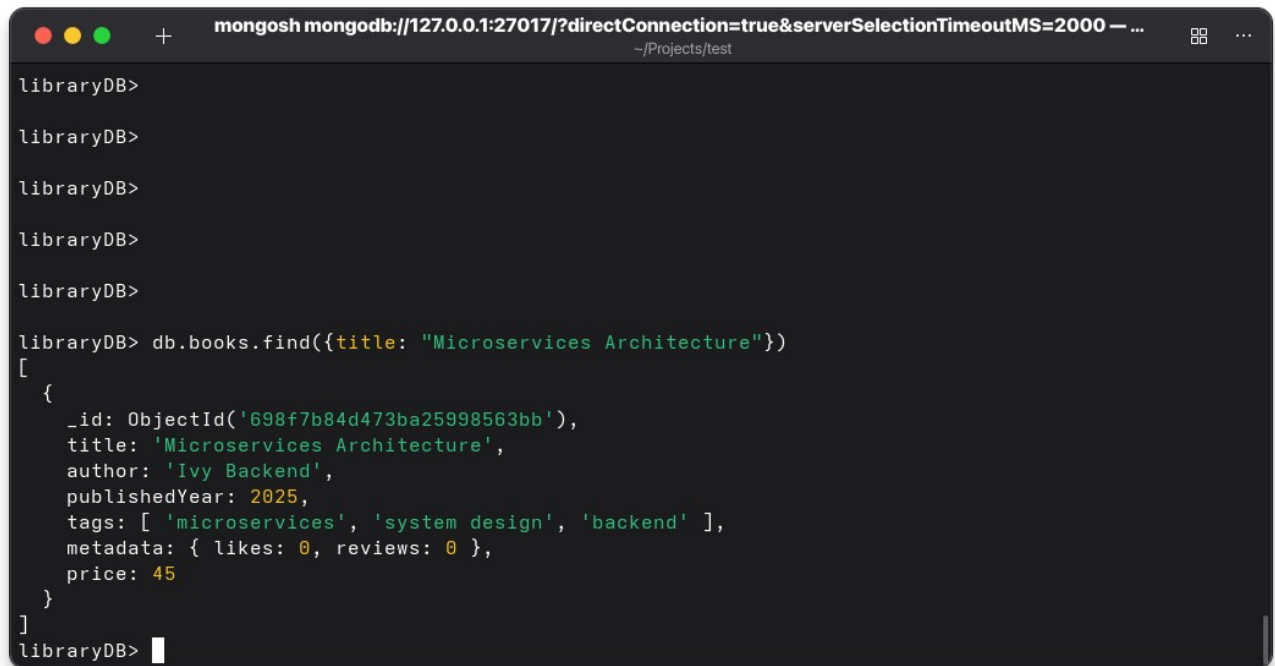


Name: Nooreldeen Ayman Mohammed Elmobashar
ITI Mansoura – Open Source
Mongo – Lab 2

1. Find a book by its title "Microservices Architecture".

A screenshot of a MongoDB terminal window. The title bar shows the connection string 'mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000' and the path '~/.Projects/test'. The terminal shows the 'libraryDB' prompt, followed by five empty prompts. Then, the command 'db.books.find({title: "Microservices Architecture"})' is entered. The output is a JSON array containing one document with fields: '_id', 'title', 'author', 'publishedYear', 'tags', 'metadata', and 'price'.

```
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.find({title: "Microservices Architecture"})
[
  {
    _id: ObjectId('698f7b84d473ba25998563bb'),
    title: 'Microservices Architecture',
    author: 'Ivy Backend',
    publishedYear: 2025,
    tags: [ 'microservices', 'system design', 'backend' ],
    metadata: { likes: 0, reviews: 0 },
    price: 45
  }
]
libraryDB> 
```

2. Find a book published in 2023 using findOne().

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.findOne({publishedYear: 2025})
{
  _id: ObjectId('698f7b84d473ba25998563bb'),
  title: 'Microservices Architecture',
  author: 'Ivy Backend',
  publishedYear: 2025,
  tags: [ 'microservices', 'system design', 'backend' ],
  metadata: { likes: 0, reviews: 0 },
  price: 45
}
libraryDB>
```

3. Find books cheaper than \$35.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.findOne({price: {$lt: 35}})
{
  _id: ObjectId('698f7b84d473ba25998563b1'),
  title: 'Learning MongoDB',
  author: 'John Developer',
  publishedYear: 2024,
  tags: [ 'mongodb', 'database', 'nosql' ],
  metadata: { likes: 120, reviews: 30 },
  price: 25.5
}
libraryDB>
```

4. Find Books that have a metadata.likes value of type number

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({"metadata.likes": {$type: "number"}}).limit(2)
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql' ],
    metadata: { likes: 120, reviews: 30 },
    price: 25.5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'nosql', 'database' ],
    metadata: { likes: 90, reviews: 12 },
    price: 30
  }
]
libraryDB> 
```

5. Find Books where tags is stored as an array

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB> db.books.find({"tags": {$type: "array"}})
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql' ],
    metadata: { likes: 120, reviews: 30 },
    price: 25.5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'nosql', 'database' ],
    metadata: { likes: 90, reviews: 12 },
    price: 30
  },
  {
    id: ObjectId('698f7b84d473ba25998563b3').
  }
]
```

6. Find Books where the discount field does NOT exist

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({discount: {$exists: 0}}).limit(2)
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql' ],
    metadata: { likes: 120, reviews: 30 },
    price: 25.5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'nosql', 'database' ],
    metadata: { likes: 90, reviews: 12 },
    price: 30
  }
]
libraryDB>
```

7. Update the price of "Microservices Architecture" to \$50.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateMany({title: "Microservices Architecture"}, {$set: {price: 50}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB>
```

8. Update the author of "Python Programming" to "Alice Code Jr.".

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateMany({title: "Python Programming"}, {$set: {author: "Alice Code Jr."}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB> 
```

9. Delete a book with title "DevOps Fundamentals".

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.deleteOne({title: "DevOps Fundamentals"})
{ acknowledged: true, deletedCount: 1 }
libraryDB> 
```

10. Delete all books published before 2021.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.deleteMany({publishedYear: {$lt: 2021}})
{ acknowledged: true, deletedCount: 1 }
libraryDB>

libraryDB>

libraryDB>

libraryDB>

libraryDB>

libraryDB>

libraryDB>

libraryDB>

libraryDB>

libraryDB>
```

11. Sort by publishedYear descending and get the first 3 books

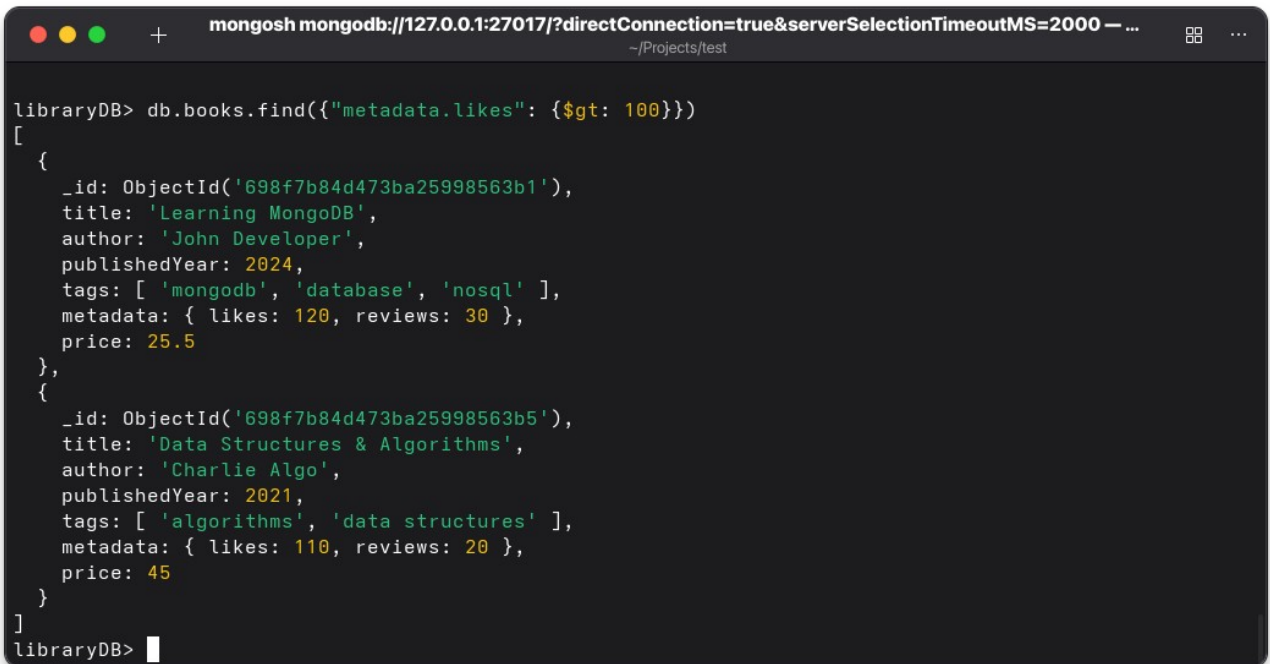
```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find().sort({publishedYear: -1}).limit(3)
[
  {
    _id: ObjectId('698f7b84d473ba25998563bc'),
    title: 'Cloud Computing Essentials',
    author: 'Jack Cloud',
    publishedYear: 2025,
    tags: [ 'cloud', 'devops', 'infrastructure' ],
    metadata: { likes: 20, reviews: 5 },
    price: 55
  },
  {
    _id: ObjectId('698f7b84d473ba25998563bb'),
    title: 'Microservices Architecture',
    author: 'Ivy Backend',
    publishedYear: 2025,
    tags: [ 'microservices', 'system design', 'backend' ],
    metadata: { likes: 0, reviews: 0 },
    price: 50
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',

```

Embedded Documents Operations

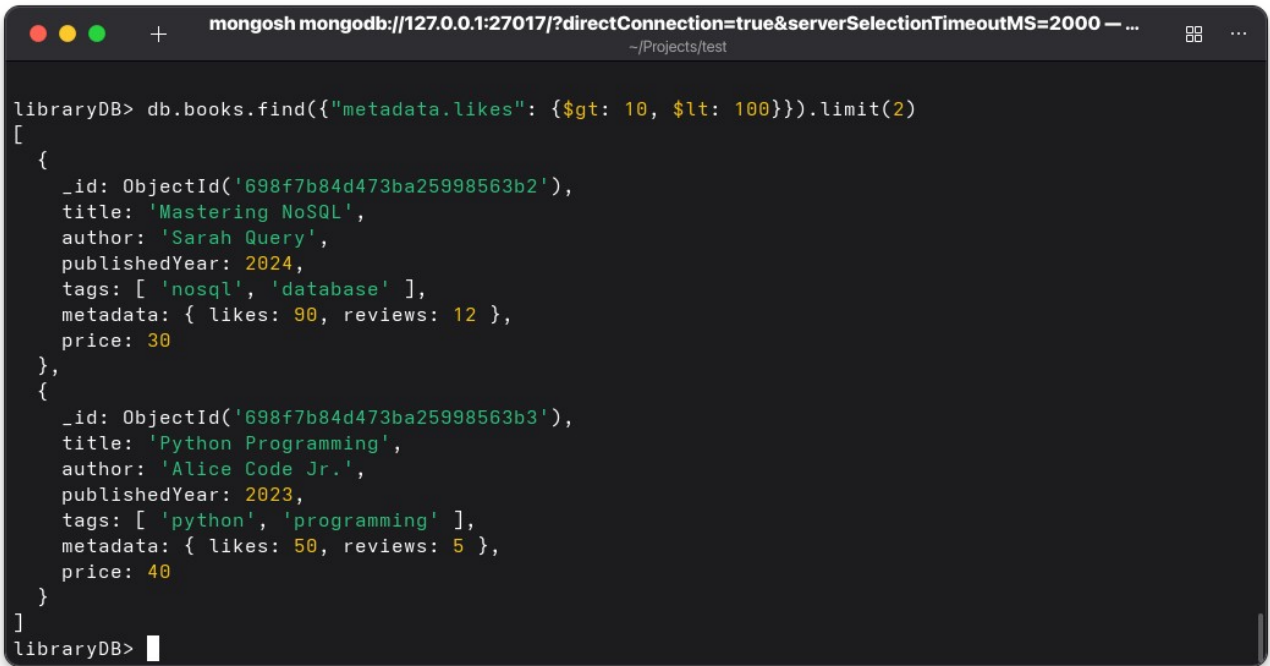
12. Find books where metadata.likes > 100.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({"metadata.likes": {$gt: 100}})
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql' ],
    metadata: { likes: 120, reviews: 30 },
    price: 25.5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b5'),
    title: 'Data Structures & Algorithms',
    author: 'Charlie Algo',
    publishedYear: 2021,
    tags: [ 'algorithms', 'data structures' ],
    metadata: { likes: 110, reviews: 20 },
    price: 45
  }
]
libraryDB>
```

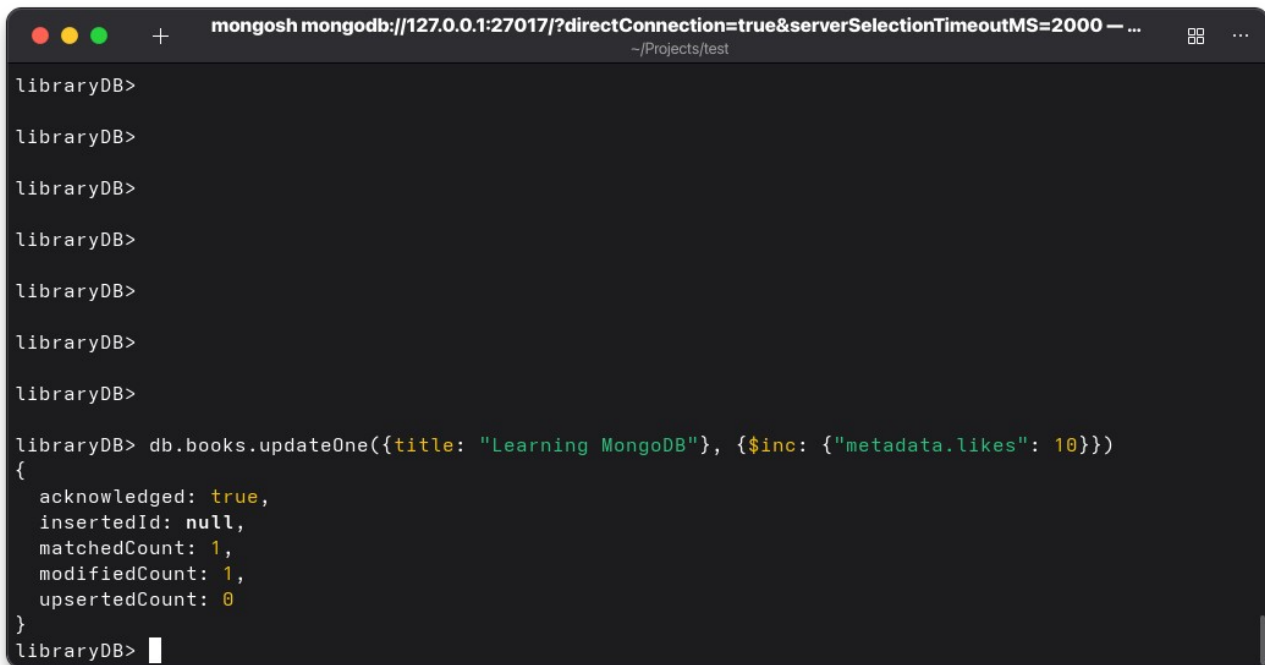
13. Find books where metadata.reviews >= 10 and metadata.likes < 100.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({"metadata.likes": {$gt: 10, $lt: 100}}).limit(2)
[
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'nosql', 'database' ],
    metadata: { likes: 90, reviews: 12 },
    price: 30
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b3'),
    title: 'Python Programming',
    author: 'Alice Code Jr.',
    publishedYear: 2023,
    tags: [ 'python', 'programming' ],
    metadata: { likes: 50, reviews: 5 },
    price: 40
  }
]
libraryDB>
```

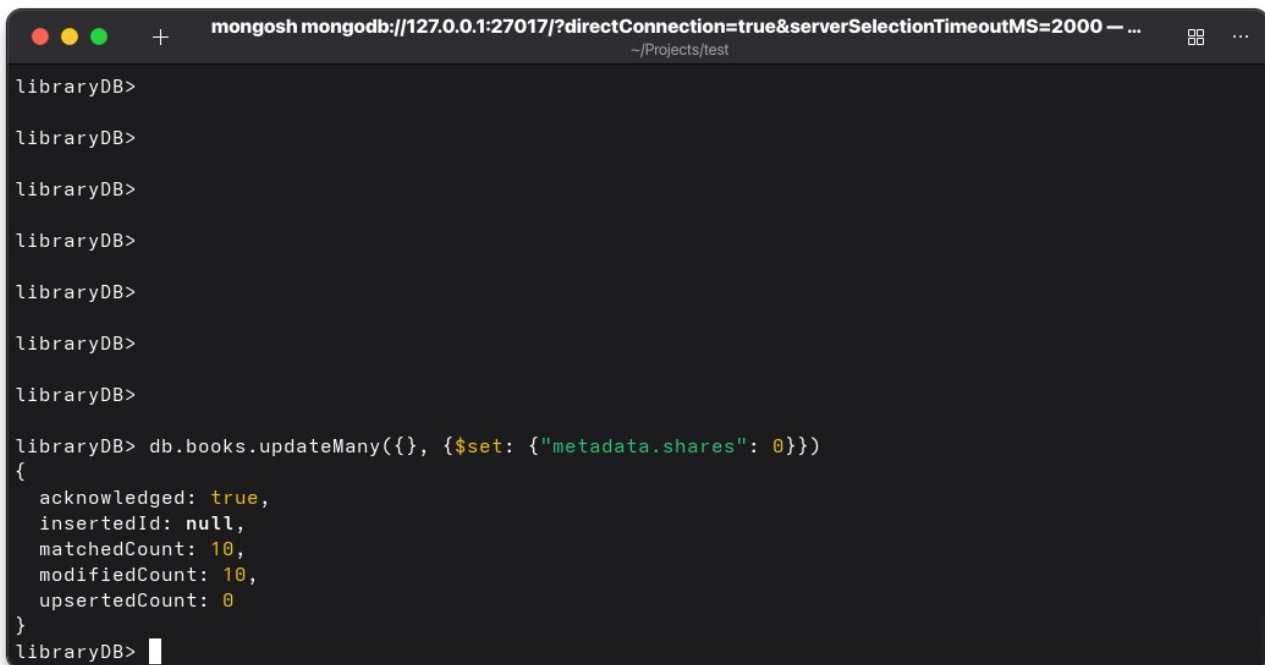
14. Increment metadata.likes of "Learning MongoDB" by 10.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateOne({title: "Learning MongoDB"}, {$inc: {"metadata.likes": 10}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB> 
```

15. Add a new field metadata.shares with value 0 to all books.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateMany({}, {$set: {"metadata.shares": 0}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 10,
  modifiedCount: 10,
  upsertedCount: 0
}
libraryDB> 
```

16. Rename metadata.reviews to metadata.reviewCount in "Advanced JavaScript".


```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.updateOne({title: "Advanced JavaScript"}, {$rename: {"metadata.reviews": "metadata.reviewCount"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB> 
```

17. Remove the metadata.likes field from "Database Design".

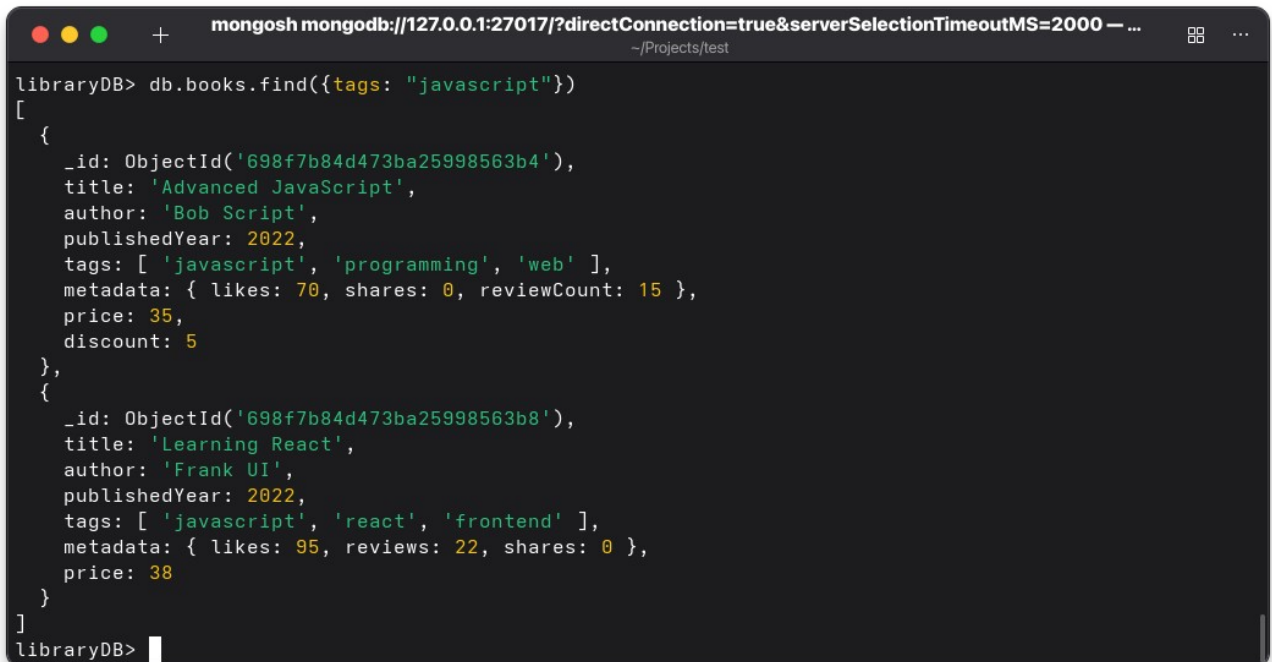
```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.updateOne({title: "Database Design"}, {$unset: {"metadata.likes": ""}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB> *
```

Array Field Operations

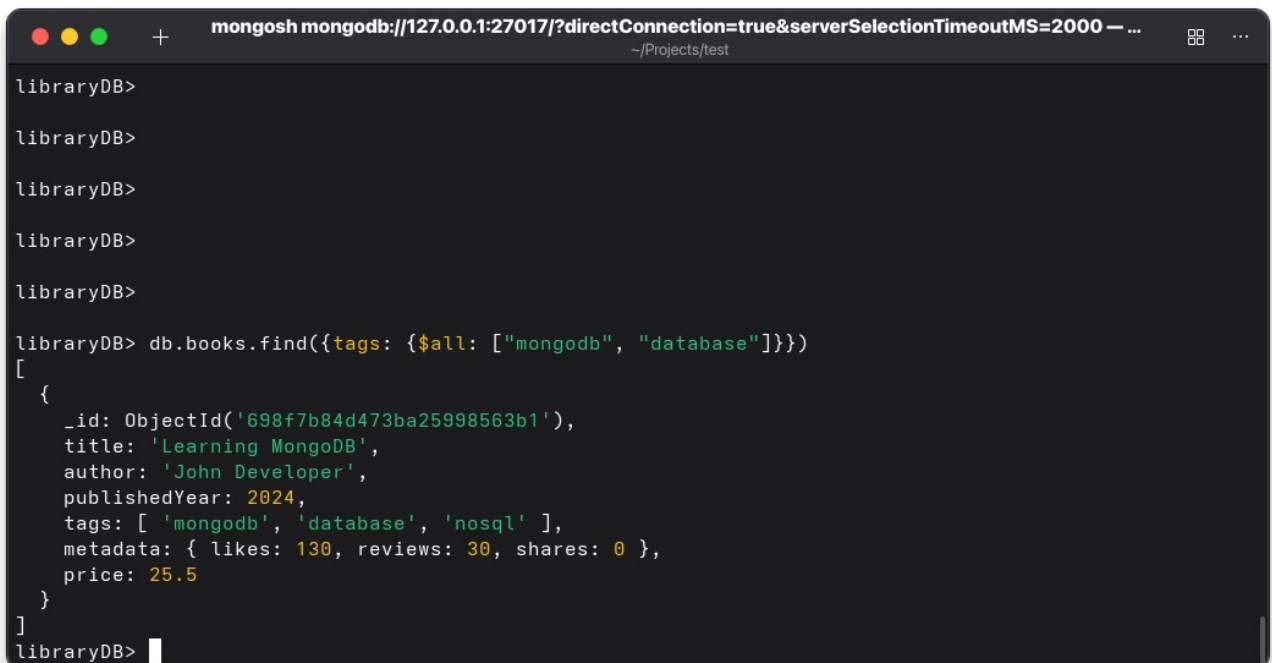
18. Find books where tags includes "javascript".



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({tags: "javascript"})
[
  {
    _id: ObjectId('698f7b84d473ba25998563b4'),
    title: 'Advanced JavaScript',
    author: 'Bob Script',
    publishedYear: 2022,
    tags: [ 'javascript', 'programming', 'web' ],
    metadata: { likes: 70, shares: 0, reviewCount: 15 },
    price: 35,
    discount: 5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b8'),
    title: 'Learning React',
    author: 'Frank UI',
    publishedYear: 2022,
    tags: [ 'javascript', 'react', 'frontend' ],
    metadata: { likes: 95, reviews: 22, shares: 0 },
    price: 38
  }
]
libraryDB>
```

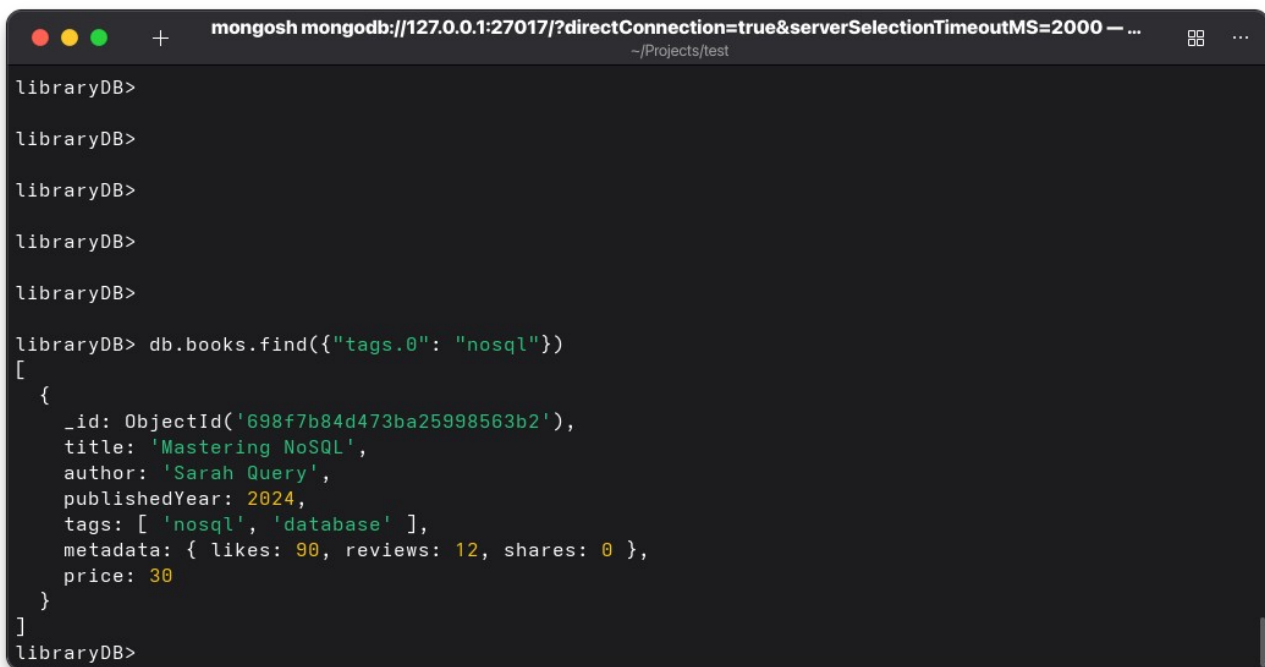
19. Find books where tags include both "mongodb" and "database".



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.find({tags: {$all: ["mongodb", "database"]}})
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql' ],
    metadata: { likes: 130, reviews: 30, shares: 0 },
    price: 25.5
  }
]
libraryDB>
```

20. Find books where the first tag is "nosql".

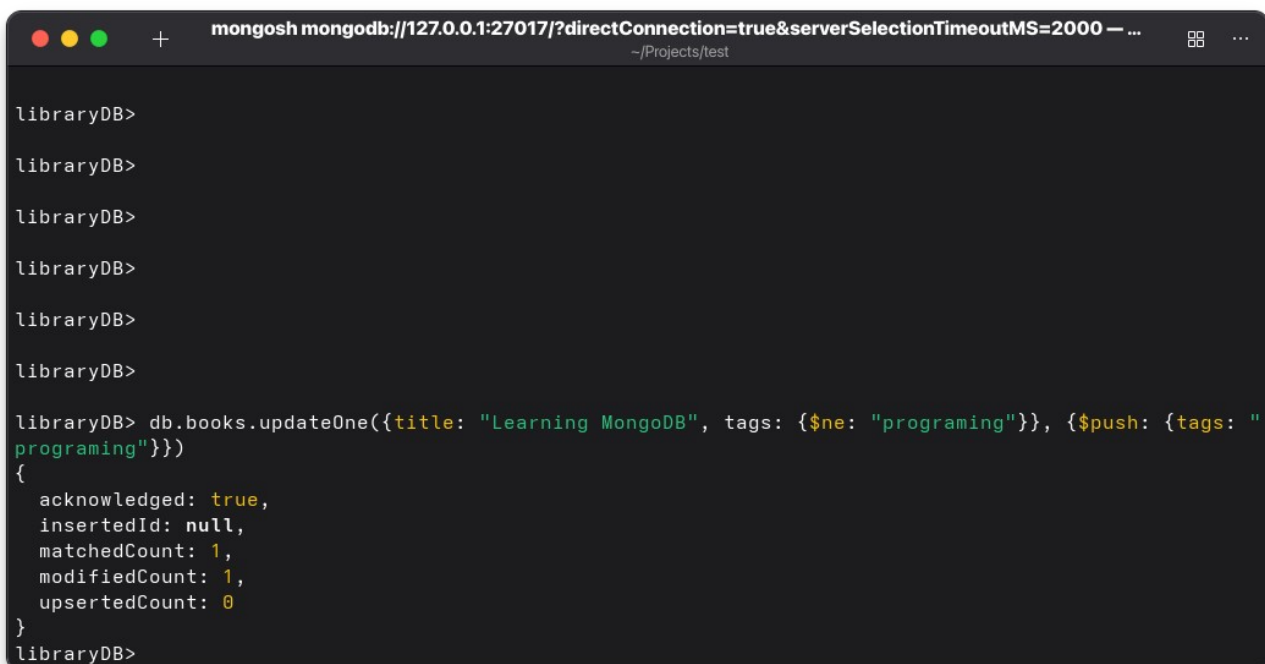


```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.find({"tags.0": "nosql"})
[
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'nosql', 'database' ],
    metadata: { likes: 90, reviews: 12, shares: 0 },
    price: 30
  }
]
libraryDB>
```

22. Add "programming" tag to "Learning MongoDB" if it doesn't exist.

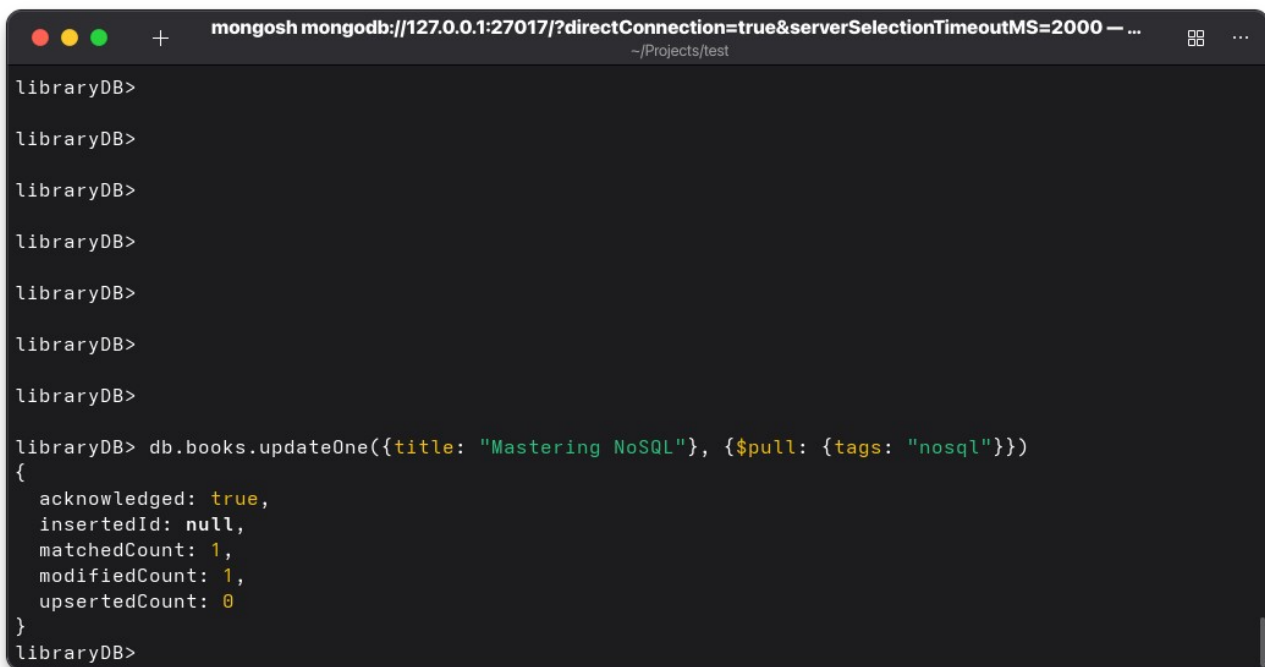


```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.updateOne({title: "Learning MongoDB", tags: {$ne: "programming"}}, {$push: {tags: "programming"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB>
```

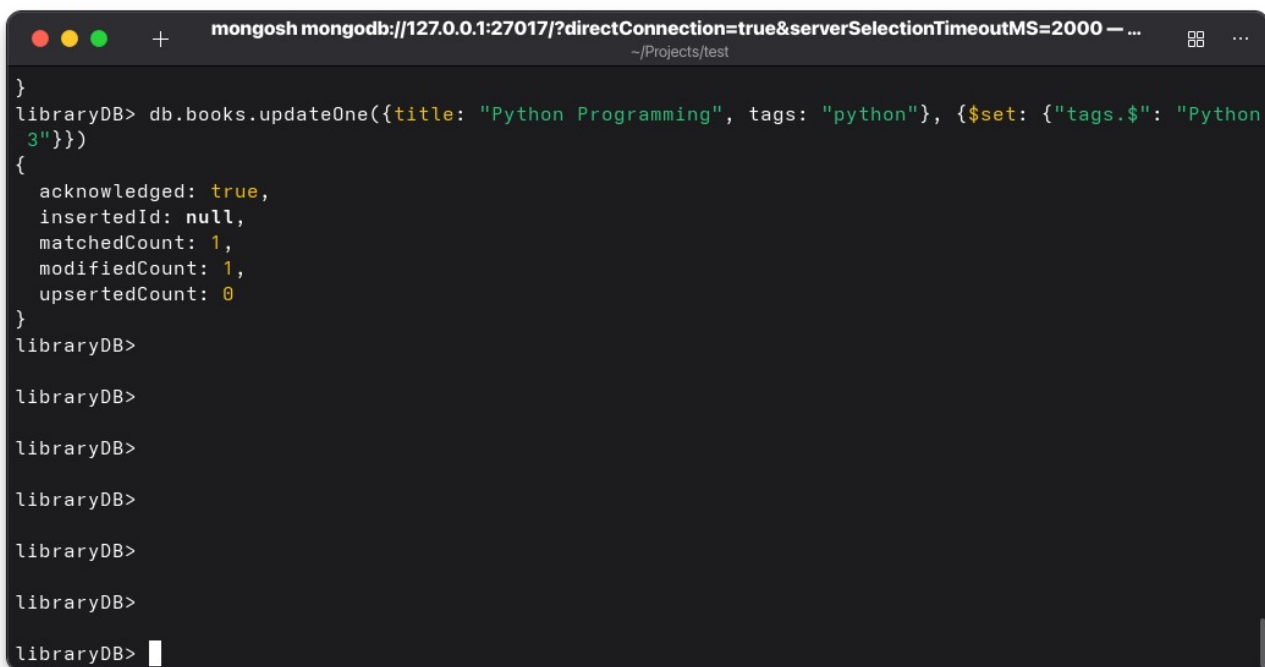
23. Remove "nosql" tag from "Mastering NoSQL".



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateOne({title: "Mastering NoSQL"}, {$pull: {tags: "nosql"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB>
```

24. Update "python" tag in "Python Programming" to "Python 3".



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

}
libraryDB> db.books.updateOne({title: "Python Programming", tags: "python"}, {$set: {"tags.$": "Python 3"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
```

25. Push a new tag "bestseller" to all books published after 2023.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.updateMany({publishedYear: {$gt: 2023}}, {$push: {tags: "bestseller"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
libraryDB> 
```

26. Pull "frontend" tag from books where it exists.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>

libraryDB> db.books.updateMany({tags: "frontend"}, {$pull: {tags: "frontend"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB> 
```

27. Add multiple tags at once: "cloud", "devops" to "DevOps Fundamentals".

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateOne({title: "DevOps Fundamentals"}, {$push: {tags: ["cloud", "devops"]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
libraryDB>
```

Combined Queries (Embedded + Array)

28. Find books where metadata.likes > 50 and tags contains "database".

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({"metadata.likes": {$gt: 50}, tags: "database"})
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql', 'programing', 'bestseller' ],
    metadata: { likes: 130, reviews: 30, shares: 0 },
    price: 25.5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'database', 'bestseller' ],
    metadata: { likes: 90, reviews: 12, shares: 0 },
    price: 30
  }
]
libraryDB>
```

29. Increment metadata.likes by 5 only for books where tags include "nosql".

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB>
libraryDB> db.books.updateMany({tags: "nosql"}, {$inc: {"metadata.likes": 5}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
libraryDB> 
```

30. Find all books where tags is an array and metadata.reviews exists.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.find({tags: {$type: "array"}, "metadata.reviews": {$exists: true}}).limit(3)
[
  {
    _id: ObjectId('698f7b84d473ba25998563b1'),
    title: 'Learning MongoDB',
    author: 'John Developer',
    publishedYear: 2024,
    tags: [ 'mongodb', 'database', 'nosql', 'programing', 'bestseller' ],
    metadata: { likes: 135, reviews: 30, shares: 0 },
    price: 25.5
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b2'),
    title: 'Mastering NoSQL',
    author: 'Sarah Query',
    publishedYear: 2024,
    tags: [ 'database', 'bestseller' ],
    metadata: { likes: 90, reviews: 12, shares: 0 },
    price: 30
  },
  {
    _id: ObjectId('698f7b84d473ba25998563b3'),
    title: 'Python Programming'
  }
]
```

31. What is \$elemMatch, \$elemMatch vs \$all?

\$elemMatch Requires One element to meet all conditions

\$all it ensures that all specified values exist somewhere within the array

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — ...
~/Projects/test

libraryDB> db.books.insertOne({
|   title: "Learning MongoDB",
|   tags: [
|     { name: "mongodb", level: "advanced" },
|     { name: "database", level: "beginner" }
|   ]
| })
{
  acknowledged: true,
  insertedId: ObjectId('698f8404d473ba25998563bd')
}
libraryDB> db.books.find({tags: {$elemMatch: {name: "mongodb", level: "advanced"}}})
[
  {
    _id: ObjectId('698f8404d473ba25998563bd'),
    title: 'Learning MongoDB',
    tags: [
      { name: 'mongodb', level: 'advanced' },
      { name: 'database', level: 'beginner' }
    ]
  }
]
libraryDB> 
```

32. what difference addToSet VS push , pop vs pull

\$push: Adds value to the array regardless of duplicates

\$addToSet: Adds the value only if it does not already exists

\$pop: Removes an element based on its position

\$pull: Removes elements based on their value or a condition